

Improvement of endothelium-dependent coronary vasodilation after a single LDL apheresis in patients with hypercholesterolemia.

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The purpose of this study was to determine whether a single LDL apheresis would improve impaired endothelium-dependent dilation of the coronary artery in patients with hypercholesterolemia. Hypercholesterolemia is associated with impaired endothelial function, and human studies using cholesterol-lowering drugs indicate that endothelial function in the coronary arteries improves with reduction of serum LDL cholesterol over 6 to 12 months. The internal diameter of the left coronary artery and the coronary blood flow were measured by intracoronary Doppler-wire measurement and quantitative angiography before and immediately after a single LDL apheresis in a population of 15 patients with familial hypercholesterolemia. Endothelium-dependent vasodilation was assessed by intracoronary infusion of acetylcholine (1, 10, and 50 microg/min), and endothelium-independent vasodilation was assessed by intracoronary bolus infusion of isosorbide dinitrate (2.5 mg) or papaverine (10 mg). A single 3-hour LDL apheresis reduced serum LDL cholesterol by an average of 86.6 +/- 1.7%. After the LDL apheresis, the changes in the coronary artery diameter and coronary blood flow in response to an infusion of 50 microg/min of acetylcholine increased significantly compared to the pre-apheresis values (from -19.7 +/- 4.8 to -2.9 +/- 3.0% [P < 0.01] and from 80.7 +/- 27.6 to 155.3 +/- 23.5% [P < 0.01], respectively). The LDL apheresis did not significantly change the response of either parameter to infusion with isosorbide dinitrate or papaverine. The endothelial function of the epicardial coronary artery and the coronary microvasculature improved in hypercholesterolemic patients after only a single LDL apheresis, a procedure that markedly reduces the serum level of LDL cholesterol. Copyright 2004 Wiley-Liss, Inc.